

CMAN :=

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Task 1 :=

$$I_{\text{col0}} = \frac{V_{\text{row0}}}{R[0][0]} = \frac{1V}{1000\Omega} = 1.0\text{mA}$$

Here No sneak path exists because every other node is Pinned to 0V.

Task 2 :=

Sneak path read :=

KCL at $V_{\text{c1}}(\text{col 1})$:=

$$\frac{(V_{\text{row0}} - V_{\text{c1}})}{R[0][1]} + \frac{(V_{\text{r1}} - V_{\text{c1}})}{R[1][1]} = 0$$

$$\frac{(1 - V_{\text{c1}})}{2000} + \frac{(V_{\text{r1}} - V_{\text{c1}})}{1000} = 0$$

Multiply through by 2000

$$(1 - V_{\text{c1}}) + 2(V_{\text{r1}} - V_{\text{c1}}) = 0$$

$$1 - V_{\text{c1}} + 2 \cdot V_{\text{r1}} - 2V_{\text{c1}} = 0$$

$$1 + 2 \cdot V_{\text{r1}} - 3 \cdot V_{\text{c1}} = 0 \leftarrow \textcircled{1}$$

KCL at $V_{\text{r1}}(\text{row 1})$:=

$$\frac{(V_{\text{r1}} - 0)}{R[1][0]} + \frac{(V_{\text{r1}} - V_{\text{c1}})}{R[1][1]} = 0$$

$$\frac{V_{\text{r1}}}{2000} + \frac{V_{\text{r1}} - V_{\text{c1}}}{1000} = 0$$

Multiply through by 2000

(2)

$$V_{-r1} + 2(V_{-r1} - V_{-c1}) = 0$$

$$V_{-r1} + 2 \cdot V_{-r1} - 2 \cdot V_{-c1} = 0$$

$$3 \cdot V_{-r1} - 2 \cdot V_{-c1} = 0 \leftarrow (2)$$

from (2) $V_{-c1} = (3/2) \cdot V_{-r1}$

Substitute into equ (1)

$$1 + 2 \cdot V_{-r1} - 3 \cdot (3/2) \cdot V_{-r1} = 0$$

$$1 + 2 \cdot V_{-r1} - 4.5 \cdot V_{-r1} = 0$$

$$1 - 2.5 \cdot V_{-r1} = 0$$

$$\boxed{V_{-r1} = 0.4 \text{ V}}$$

Then $V_{-c1} = (3/2) \times 0.4 = 0.6 \text{ V}$

$$\boxed{V_{\text{row } 1} = 0.4 \text{ V}, \quad V_{\text{col } 1} = 0.6 \text{ V}}$$

So, through $R[1][0]$ from the floating row 1
which is 0.4 V to Col 0.

$$I_{\text{Sneak}} = \frac{(0.4 - 0)}{2000} = 0.2 \text{ mA}$$

$$\boxed{I_{\text{col } 0} (\text{actual}) = 1.0 \text{ mA} + 0.2 \text{ mA} = 1.2 \text{ mA}}$$

Task 3 :=

When we read a column, extra current sneaks in through floating rows and columns that we never intended to use like the 0.2mA we found in Task 2 (I-sneak). This extra current makes the output higher than the correct MVM answer it gives us the wrong results. In a large array this gets much worse because hundreds of these unwanted paths all dump extra current at the same time making the whole calculation unreliable.